

Phase 5 TSP Runbook

This runbook covers the benchmark-transfer phase that moves the replay-aware curriculum study onto a real combinatorial-optimization family.

1. Prepare The Benchmark Pack

Activate the environment first:

```
conda activate Python3_14
python prepare_tsp_benchmarks.py
```

That downloads the selected TSPLIB95 EUC_2D subset, computes best-known tour costs from the official .opt.tour files, and writes benchmarks/tsp/manifest.json.

2. Offline Smoke Test

```
python run_tsp_suite.py --config configs/tsp_suite/smoke_suite.json --skip-judge
```

Use this first whenever you change the TSP engine, benchmark loader, or replay-selection logic. The smoke suite writes into DO NOT COMMIT/tsp_suite/smoke because it is a verification artifact, not an official experiment run.

3. Main Phase-5 Suite

```
python run_tsp_suite.py --config configs/tsp_suite/01_replay_transfer.json
```

The four default conditions are:

- tsplib_no_replay
- tsplib_random_replay
- tsplib_failure_replay
- tsplib_failure_replay_compression

Primary interpretation target:

- final held-out TSPLIB optimality gap

Secondary targets:

- final synthetic holdout gap
- combined transfer gap
- code novelty
- heuristic complexity
- adaptation efficiency

Selection-time transfer probes combine held-out TSPLIB probe instances with the adversarial geometric layout pool. The exact synthetic holdouts remain reserved for final evaluation.

4. Replication

Use paired seed offsets exactly as in the earlier grid phases:

```
python run_tsp_suite.py --config configs/tsp_suite/01_replay_transfer.json --seed-offset 1000
--replicate-label b
```

Recommended minimum:

- 10 paired offsets for an official result

Official main-study target:

- 20 paired offsets using the fixed offset list 0, 1000, 2000, ..., 19000
- Use the same offset list for every condition and every routing family so the paired comparisons remain valid.

5. Aggregation

```
python aggregate_tsp_runs.py --runs-root runs/tsp_suite/replay_transfer
```

That writes:

- aggregate_summary.json
- aggregate_report.md
- aggregate_report.pdf
- paired bootstrap comparison tables for the replay conditions

6. Interpretation Rules

- Do not treat replay probes or transfer probes inside training as the final evidence; final held-out evaluation remains the endpoint.
- Do not treat code novelty as algorithmic invention without corresponding transfer improvements.
- The compression-aware replay condition matters only if it matches or improves transfer while reducing novelty or complexity growth.